There may have been a time when sense organs communicated more or less directly with muscles; indeed, sea anemones are not far from this state today, since for their way of life it is efficient. But to achieve more complex and indirect relationships between the timing of events in the outside world and the timing of muscular contractions, some kind of brain was needed as an intermediary. A notable advance was the evolutionary 'invention' of memory. By this device, the timing of muscle contractions could be influenced not only by events in the immediate past, but by events in the distant past as well. The memory, or store, is an essential part of a digital computer too. Computer memories are more reliable than human ones, but they are less capacious, and enormously less sophisticated in their techniques of information-retrieval.

One of the most striking properties of survival-machine behaviour is its apparent purposiveness. By this I do not just mean that it seems to be well calculated to help the animal's genes to survive, although of course it is. I am talking about a closer analogy to human purposeful behaviour. When we watch an animal 'searching' for food, or for a mate, or for a lost child, we can hardly help imputing to it some of the subjective feelings we ourselves experience when we search. These may include 'desire' for some object, a 'mental picture' of the desired object, an 'aim' or 'end in view'. Each one of us knows, from the evidence of our own introspection, that, at least in one modern survival machine, this purposiveness has evolved the property we call 'consciousness'. I am not philosopher enough to discuss what this means, but fortunately it does not matter for our present purposes because it is easy to talk about machines that behave *as if* motivated by a purpose, and to leave open the question whether they actually are conscious. These machines are basically very simple, and the principles of unconscious purposive behaviour are among the commonplaces of engineering science. The classic example is the Watt steam governor.

The fundamental principle involved is called negative feedback, of which there are various different forms. In general what happens is this. The 'purpose machine', the machine or thing that behaves as if it had a conscious purpose, is equipped with some kind of measuring device which measures the discrepancy between the current state of things, and the 'desired' state. It is built in such a way that the larger this discrepancy is, the harder the machine works. In this way the machine will automatically tend to reduce the discrepancy—this is why it is called